Lecture 86

Coevolution:

The evolution of ecologically related species is sometimes coordinated such that each species exerts a strong selective influence on the other. This is **Coevolution.** Coevolution may occur when species are competing for the same resource or during predator–prey interactions.

When populations of 2 different species interact, then changes in the gene pool of one species can lead to changes in the gene pool of the other species. Such changes can help both sides to become more competitive or can help to avoid/reduce competition. Biologists call this **Coevolution.**

One example of Coevolution is the relationships b/w some flowering plants & their animal pollinators. **Insect**-pollinated flowers are usually **yellow or blue** because insects see these wavelengths of light best.

Flowers pollinated by **hummingbirds**, are often **tubular & red.** Hummingbirds have a poor sense of smell but see red very well. The long beak of hummingbirds is an adaptation that allows them to reach far into tubular flowers.



Long-term interactions b/w **bats & their prey** such as **moths and butterflies** can lead to Coevolution, as the bats evolve traits that increase their chances of getting a meal & the moths evolve traits that help them avoid being eaten.

Lecture 87

Symbiosis:

Some of the best examples of adaptations arising through Coevolution come from two different species living in continuing, intimate associations, called **symbiosis (Gr. *sym, together bio, life).***

Symbiosis ('living together') is the term that has been coined for such close physical associations b/w species, in which a 'symbiont' occupies a habitat provided by a 'host'.

In some instances, one member of the association benefits, and the other is harmed. In other cases, life without the partner would be impossible for both.

Common form of symbiosis are

- **Commensalism**
- **Mutualism**

Lecture 88

Commensalism:

Commensalism is a symbiotic relationship in which one member of the relationship benefits, and the second is neither helped nor harmed.

In tropical forests certain kinds of silverfish insects move along with columns of army ants to share the food obtained by the ants in their raids. The army ants receive no apparent harm or benefit from the silverfish.



Epiphytic plants such as certain types of **orchids & bromeliads** attach themselves to trunks or branches of large trees in tropical & subtropical forests. These air plants benefit by getting a solid base on which they grow.

They also live in an elevated spot that gives them better access to sunlight, water from the humid air & rain, & nutrients falling from the tree's upper leaves & limbs. Their presence apparently does not harm the tree.



A **clownfish** gains protection & food by living among deadly stinging **sea anemones** & helps protect the anemones from some of their predators.

Some other forms of mutualism are,

- **Lichen**
- **Mycorrhizae**
- **Root nodules**

Lecture 89

Mutualism:

Mutualism is a symbiotic relationship that benefits both members. In mutualism, two species behave in ways that benefit both by providing each with food, shelter, or some other resource.

For example, **honeybees, caterpillars, butterflies**, & other insects feed on a male flower's nectar, picking up pollen in the process, & then pollinating female flowers when they feed on them.

In gut inhabitant mutualism, vast armies of bacteria in the digestive systems of animals help to break down (digest) their hosts' food. In turn, the bacteria receive a sheltered habitat and food from their host.



Oxpeckers or tickbirds feed on parasitic ticks that infest large, thick skinned animals such as the endangered **black rhinoceros**.

Lecture 90

Lichens:

A mutualistic association b/w **algae** & **fungus** in which both the partners get benefits from each other is known as **lichen**. Fungus are usually **Ascomycetes** or **Basidiomycetes** while the other partner is **algae or cyanobacteria**.

Growth forms of Lichens

1. Crustose
2. Foliose
3. Fruticose



Crustose:

A type of lichen that form a flat crust on the substratum & firmly attached to substratum.



Foliose:

This type of lichens have flat leaf like structures & are attached to substratum by thread like structures.



Fruticose:

This type of lichens have erect filamentous like outgrowths & possess vertical pattern of growth.

Benefits of Lichens:

- Bio indicators
- Ecosystem sustainability
- Medicines
- Animal food
- Dye for clothes



Ectomycorrhizas:

Ectomycorrhizal fungi form symbioses with many trees and shrubs, dominating boreal and temperate forests and also some tropical rainforests.

Ericoid mycorrhizas:

Ericoid mycorrhizas are found in the dominant species of heathlands (barren infertile lands) including the northern hemisphere heaths and heathers (**Ericaceae**) and the Australian heaths (**Epacridaceae**).

Lecture 91

Mycorrhizae:

A mutualistic relationship b/w fungi & roots of higher plants by which both partners are get benefited. A very remarkable group of **Ascomycetes** fungi, **Clavicipitaceae**, grow in tissues of many species of grass.

The fungal networks in Mycorrhizae capture minerals from the soil & transport them to the plants while in turn plants provides nutrients like sugars to fungus.



Major types of Mycorrhizae

1. Endomycorrhizas or Arbuscular mycorrhizas
2. Ectomycorrhizas
3. Ericoid mycorrhizas

Endomycorrhizas or Arbuscular mycorrhizas:

They are found in up to two-thirds of all plant species, including most nonwoody species & tropical trees. Fungus incorporates into cortical cells of roots of higher plant.

Lecture 92

Root nodules:

It is a mutualistic association b/w nitrogen fixing bacteria & roots of plants of legume family. The process of nitrogen fixation is done by different kinds of microorganisms.

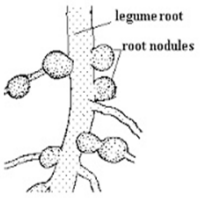
Types of nitrogen fixing bacteria

Azobactor

- Free living aerobic bacteria

Clostridium

- Free living anaerobic bacteria



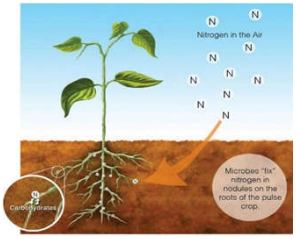
Chlorobium

- Photosynthetic bacteria

Thiobacillus

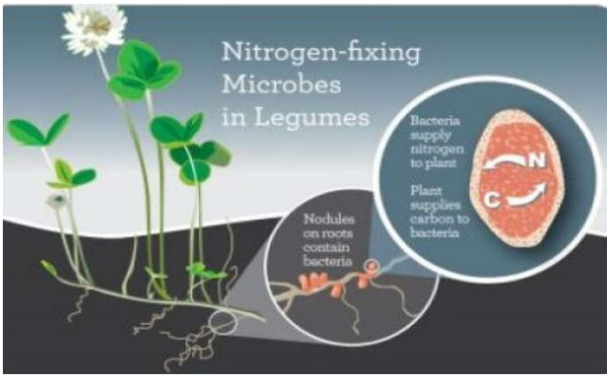
- Chemosynthetic bacteria

Nostoc - Blue green algae



A special vascular system develops in the host, supplying the products of photosynthesis to the nodule tissue & carrying away fixed nitrogen compounds to other parts of the plant. The nodule tissue provides shelter & carbon to bacteria.

This whole picture summarizes the process of nitrogen fixation.



Healthy tree on the left & an unhealthy one on the right, which is infested with parasitic **mistletoe**.

Blood-sucking parasitic sea **lampreys** attached to an adult lake trout.

The definitive or final host is the host that harbors the sexual stages of the parasite. A fertile female in a definitive host may produce thousands of eggs in its lifetime. Each egg gives rise to an immature stage that may be a parasite of a second host. This second host is called an **intermediate host**, & asexual reproduction may occur in this host.

Lecture 93

Parasitism:

Parasitism is a kind of association in which one organism called parasite lives on the expense of other organism, called a host. The host usually survives at least long enough for the parasite to complete one or more life cycles.

Unlike the typical predator, a parasite usually is much smaller to its host & rarely kills its host. Most parasites remain closely associated with their hosts. They draw nourishment from host, & may gradually weaken them over time.

Endoparasite: Some parasites live inside the body of their hosts & are named as endoparasite.

❖Pathogens

❖*Entamoeba histolytica*

❖Tapeworms

❖Hookworms



Ectoparasite: Parasites that attach themselves to the outsides of their hosts are called Ectoparasite.

• Mosquitoes

• Mistletoe plants

• Sea lampreys



Lecture 94

Interspecific adaptations:

Predators exert selective pressure on their prey, leading them to have better adaptations to survive & evade capture. In turn, the predators themselves evolve to become better hunters.

This back-and-forth evolutionary process b/w predator & prey generally leaves both species more refined.

By gaining certain kinds of adaptations predator & prey both become more active in their responses.

Common Interspecific adaptations,

- Camouflage
- Counter-shading
- Aposematic coloration
- Mimicry
- Chemical warfare

Lecture 95

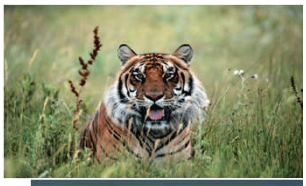
Camouflage:

Camouflage is a kind of coloration or patterns that help an animal to appear to blend with its surroundings.

Many prey species use the camouflage (the ability to change color) of certain shapes or colors or (**chameleons and cuttlefish**).

Cryptic coloration (L. *crypticus, hidden)* *is a* type of camouflage that occurs when an animal takes on color patterns in its environment to prevent the animal from being seen by other animals.

The color pattern of this tiger (***Panthera tigris***) provides effective camouflage that helps when stalking prey.



Praying **mantises** sit in flowers of a similar color & **ambush** visiting insects.



Some insect species have shapes that make them look like twigs ,bark, thorns, or even bird droppings on leaves.



A leaf insect can be almost invisible against its background, as can an arctic hare in its white winter fur.



Lecture 96

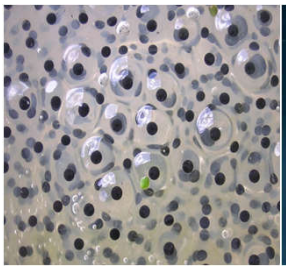
Counter-shading:

Counter shading is a type of color pattern in animals in which there is a dark pigmentation on the upper surface of body & light pigmentation on the lower surface of body.

Counter shading helps an animal to blend in with its surrounding environment. Counter shading is a kind of camouflage common in frog & toad eggs.

These eggs are darkly pigmented on top & lightly pigmented on the bottom. When a bird or other predator views the eggs from above, the dark of the top side hides the eggs from detection against the darkness below.

On the other hand, when fish view the eggs from below, the light undersurface blends with the bright air-water interface.





Penguins are another example that exhibit phenomenon of counter shading.

Lecture 97

Aposematic coloration:

Some animals protect themselves by being dangerous or distasteful to predators by using prominent coloration. These color patterns are known as **warning** or **aposematic coloration.**

The sharply contrasting white stripes of a skunk & bright colors of poisonous snakes give similar messages.





They flash a warning: “Eating me is risky.” Examples are brilliantly colored poisonous frogs is another example.

when a bird such as a blue jay eats a foul-tasting monarch butterfly it usually vomits & learns to avoid them.





Biologist **Edward O. Wilson** Described two rules, based on coloration. These rules evaluate possible danger from an unknown animal species we encounter in nature.

- First, if it is small and strikingly beautiful, it is probably poisonous.
- Second, if it is strikingly beautiful and easy to catch, it is probably deadly.

Lecture 98

Mimicry:

Mimicry (L. *mimus,* to imitate) occurs when a species resembles one, or sometimes more than one, other species & gains protection by the resemblance.

These six species of ***Heliconius*** are all distasteful to bird predators. A bird that consumes any member of the six species is likely to avoid all six species in the future.



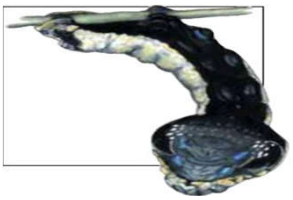
Some butterfly species, such as the non-poisonous viceroy, gain protection by looking & acting like the monarch, a protective device known as mimicry.

Viceroy butterfly mimics monarch butterfly.

Some moths have wings that look like the eyes of much larger animals. Hind wings of Io moth resemble eyes of a much larger animal.



When touched, snake caterpillar changes shape to look like head of snake.



Lecture 99

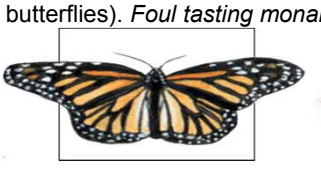
Chemical warfare:

Chemical warfare is another common strategy to escape from predators. Chemical warfare refers to use of chemical substance as a weapon for defence purpose.

Some predators use chemical warfare to attack their prey. For example, spiders and poisonous snakes use venom to paralyze their prey & to deter their predators.

Some prey species discourage predators with chemicals that are poisonous (oleander plants) & irritating (stinging nettles & bombardier beetles). ***Bombardier beetle***

Some possess foul smelling (skunks, skunk cabbages, & stinkbugs), or bad tasting (buttercups & monarch butterflies). *Foul tasting monarch butterfly.*





When attacked, some species of squid and octopus emit clouds of black ink, allowing them to escape by confusing their predators.

Lecture 100

Biome:

A large regional community is known as biome. Each biome is differentiated from one another on the base of, Plants, Animals, Climate.

Main elements of a biome are,

- ❖ Habitat
- ❖ Biodiversity
- ❖ Climate
- ❖ Plant & animal adaptation
- ❖ Human activity

There are two broad categories of biomes,

1. **Aquatic biomes**
2. **Terrestrial biomes**

Aquatic biomes: Aquatic biomes are classified into,

1-Saltwater biome, 2-Freshwater biome

Terrestrial biomes:

-Tundra biome

-Desert biome

-Grassland biome

-Forest biome

Lecture 101

Biogeochemical cycles:

The process that involves the cyclic movement of nutrients b/w biotic (living portion) & abiotic (non living) environment of earth is termed as biogeochemical cycle.

Matter is constantly recycled within ecosystems.

A nutrient is any element essential for life. Approximately 97% of living matter is made of oxygen, carbon, nitrogen, & hydrogen.

Biogeochemical Cycles are of two main types

1.Gaseous cycles 2.Sedimentary cycles

Gaseous Cycles:

Carbon cycle

Nitrogen cycle

Oxygen cycle

Hydrogen cycle

Sedimentary Cycles:

Sulfur cycle

Phosphorus cycle

Calcium cycle

These elements are less abundant than those of gases.

In Sedimentary cycles the nonliving reservoir for these nutrients (P, N, Ca.) is the earth, & cycles involving these elements are called **sedimentary cycles.**

Nutrient cycles connect past, present, & future forms of life. Some of the C atoms in your skin may once have been part of an oak leaf, a dinosaur's skin, or a layer of limestone rock.

1. **Carbon cycle**
2. **Nitrogen cycle**
3. **Phosphorus cycle**
4. **Sulfur cycle**
5. **Water cycle**

Lecture 102

Carbon cycle:

Carbon is the basic building block of the carbohydrates, fats, proteins, DNA, & other organic compounds necessary for life. It circulates through the biosphere, atmosphere, & parts of hydrosphere, in the **Carbon cycle**.

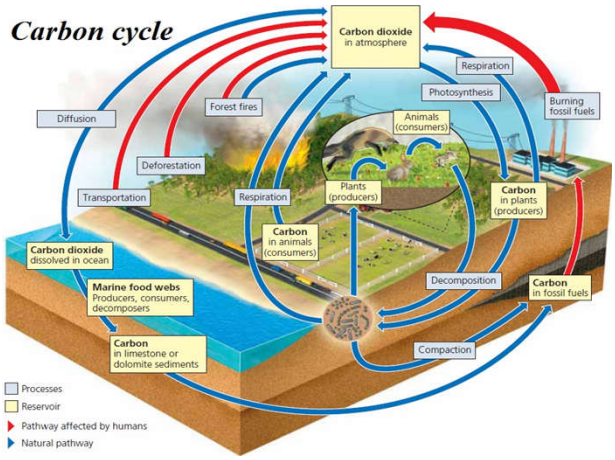
The carbon cycle is based on (CO_2) gas, which makes up **0.038%** of the volume of the atmosphere & is also dissolved in Water. CO_2 is a key component of nature's thermostat.

It depends on photosynthesis & aerobic respiration by the earth's living organisms. It is released into atmosphere by the process of aerobic respiration while consumed by the process of photosynthesis.

Terrestrial producers remove CO2 from the atmosphere and aquatic producers remove it from the water.

Photosynthesis converts CO2 into complex carbohydrates such as glucose (C6H12O6).

This linkage b/w photosynthesis in producers & aerobic respiration in producers, consumers, & decomposers circulates carbon in the biosphere. Oxygen and hydrogen, the other elements in carbohydrates, cycle almost in step with carbon.



Lecture 103

Nitrogen cycle:

The major reservoir for N2 is the atmosphere. Chemically unreactive (N2) gas makes up **78%** of the volume of the atmosphere. Nitrogen is a crucial component of proteins, many vitamins, & nucleic acids such as DNA. However, N2 cannot be absorbed & used directly as a nutrient by multicellular plants or animals.

Steps in nitrogen cycle;

- Fixation
- Nitrification
- Assimilation
- Ammonification
- Denitrification

Fixation: Two natural processes convert or fix N2 into compounds useful as nutrients for plants & animals.

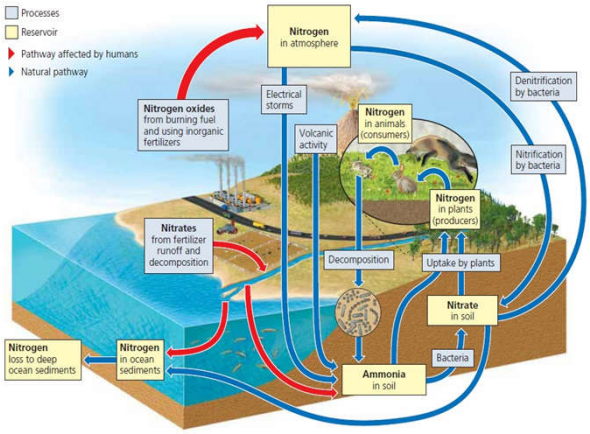
- Electrical discharges/lightning, taking place in the atmosphere.
- Nitrogen fixation by nitrogen-fixing bacteria.

Nitrification: Specialized soil bacteria convert most of the *(NH3)* & *(NH_4^+)* in soil to *(NO_3^-)* which are easily taken up by the roots of plants.

Assimilation: The plants then absorb nitrates from the soil into their roots to produce various amino acids, proteins, nucleic acids, & vitamins.

Ammonification: When a plant or animal dies, specialized decomposer bacteria convert this detritus into simpler nitrogen-containing inorganic compounds such as *(NH3)* & water-soluble salts containing *(NH_4^+)*.

Denitrification: Specialized bacteria convert *(NH3)* and *(NH_4^+)* back into nitrite and nitrate ions, & then into *(N2)* gas & *(N2O)*. These gases are released to the atmosphere to begin the nitrogen cycle again.



Phosphate can be lost from the cycle when it washes from the land into streams & rivers & is carried to the ocean. In oceans it can be deposited as marine sediment & remain trapped for millions of years.

Someday, geological processes may uplift and expose these seafloor deposits, from which phosphate can be eroded to start the cycle again.

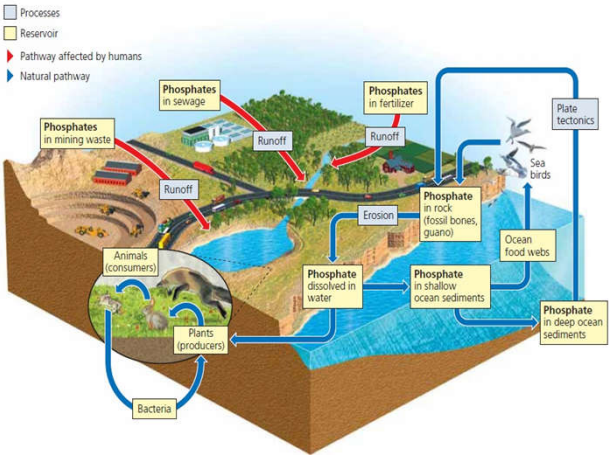
Because most soils contain little phosphate, it is often the limiting factor for plant growth on land unless phosphorus (as phosphate salts mined from the earth) is applied to the soil as an inorganic fertilizer.

Lecture 104

Phosphorus cycle:

Phosphorus circulates through water, the earth's crust, & living organisms in the **phosphorus cycle.** In contrast to the cycles of water, C, & N, the P cycle does not include the atmosphere.

The major reservoir for phosphorous is phosphate salts containing phosphate ions *($PO4^{-3}$)* in terrestrial rock formations & ocean bottom sediments. The phosphorus cycle is slow compared to the water, C, & N cycles.



Phosphorous is a component of biologically important molecules such as,

- Nucleic acids
- Energy transfer molecules (ADP and ATP)
- Vertebrate bones and teeth.

As water runs over exposed phosphorus containing rocks, it slowly erodes away inorganic compounds that contain phosphate ions *($PO4^{-3}$)*. The dissolved phosphate can be absorbed by the roots of plants and by other producers.

Lecture 105

Sulfur cycle:

Sulfur circulates through the biosphere in the **sulfur cycle.** Much of the earth's sulfur is stored underground in rocks & minerals, including sulfate *($SO4^{2-}$)* salts buried deep under ocean sediments.

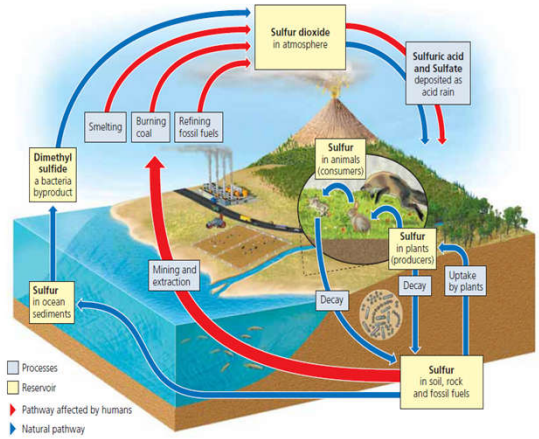
Sulfur also enters the atmosphere from several natural sources. Hydrogen sulfide (a colorless, highly poisonous gas with a rotten-egg smell) & Sulfur dioxide (a colorless & suffocating gas) is released from active volcanoes.

Particles of sulfate ($SO4^{2-}$) salts, such as ammonium sulfate, enter the atmosphere from sea spray, dust storms, and forest fires. Plant roots absorb sulfate ions & incorporate the sulfur as an essential component of many proteins.

Certain marine algae produce large amounts of volatile **DMS** (dimethyl sulfide). In the atmosphere, **DMS** is converted to sulfur dioxide, some of which in turn is converted to sulfur trioxide gas and to tiny droplets of sulfuric acid (H_2SO_4).

In the oxygen-deficient environments of flooded soils, freshwater wetlands, and tidal flats, specialized bacteria convert sulfate ions to sulfide ions.

The sulfide ions can then react with metal ions to form insoluble metallic sulfides, which are deposited as rock, and the cycle continues.



Lecture 106

Water cycle:

The **hydrologic cycle,** or **water cycle,** collects, purifies**,** & distributes the earth's fixed supply of water. Water is an amazing substance which makes the water cycle critical to life on earth.

The water cycle is a global cycle because there is a large reservoir of water in the atmosphere as well as in the hydrosphere, especially the oceans.

The water cycle is powered by energy from the sun & involves three major processes,

1- Evaporation 2- Transpiration 3- Precipitation

Evaporation: Evaporation changes water into water vapor in the atmosphere. Incoming solar energy causes evaporation of H2O from oceans, lakes, rivers, & soil.

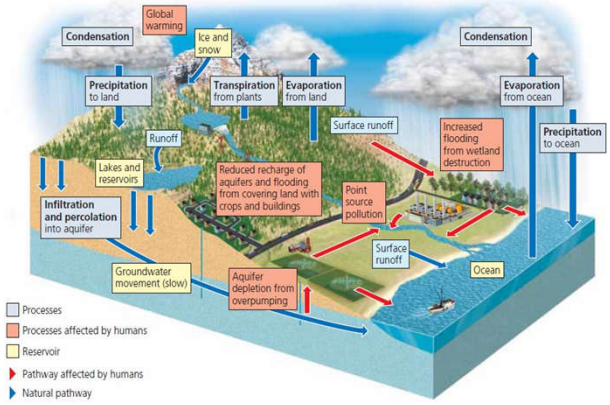
About 84% of water vapor in the atmosphere comes from the oceans, which cover almost three-fourths of the earth's surface; the rest comes from land.

Transpiration: Over land, about 90% of the water that reaches the atmosphere evaporates from the surfaces of plants through a process called **transpiration.**

Precipitation: Precipitation returns the water back to earth's surface. Precipitation becomes surface runoff that flows into streams & lakes & finally into the oceans. From oceans it can evaporate to repeat the water cycle. Some precipitation is changed to ice that is stored. Some precipitation sinks to underground layers of rock, sand, & gravel called aquifers, & stored as groundwater.

A small amount of the earth's H20 is absorbed by roots of plants. Some of this H2O is used to make food for consumers by photosynthesis, while most of H2O is evaporated through process of transpiration.

Consumers get their water from their food/by drinking it.



- The exhaust pipe of an auto mobile.



Non-point sources are dispersed & often difficult to identify. Examples are pesticides blown from the land into the air & the runoff of fertilizers & pesticides from farmlands, lawns, gardens, & golf courses into streams & lakes.

It is much easier and cheaper to identify and control or prevent pollution from point sources than from widely dispersed nonpoint sources.

Lecture 107

Pollution:

Pollution is anything in the environment that is harmful to the health, survival, or activities of humans or other organisms. Pollution is any detrimental change in an ecosystem.

Most kinds of pollution are the results of human activities. Large human populations & demands for increasing goods & services contribute to pollution problems.

There are two main types of pollutants.

- **Biodegradable pollutants**
- **Non-degradable pollutants**

Biodegradable pollutants are harmful materials that can be broken down by natural processes. Examples are human sewage and newspapers.

Non-degradable pollutants are harmful materials that natural processes cannot break down. Examples are toxic chemical elements such as Pb, Hg, & As.

Pollutants can enter the environment naturally through,

Volcanic eruptions-Human activities

-Burning coal and gasoline

-Discharging chemicals into rivers & the ocean.

The pollutants come from two types of sources.

1. **Point sources**
2. **Non point sources**

Point sources are single, identifiable sources. e.g; are,

- Smokestack of a coal-burning power /industrial plant.
- The drainpipe of a factory,

Lecture 108

Organic pollutants: Bio stimulants:

Organic pollutants or biostimulants, primarily from agricultural fertilizers & sewage waste, have a major impact on aquatic environments. When these excessive nutrients enter aquatic systems, they stimulate plant growth.

Rapid phytoplankton growth or algal blooms creates **hypoxic** (that is, **low-oxygen, less than 2 mg/liter**) or even **anoxic** (**no oxygen**) environments in which few organisms can survive.

Large concentrations of algae also reduce water clarity, preventing light from reaching the bottom & reducing the growth of seagrasses.

Eutrophication:

A long-term increase in excess nutrients into an ecosystem causes higher biological activity this is known as eutrophication or harmful algal blooms.

It decreases the oxygen content rapidly.

Effects of water pollution

- Food chain imbalance
- Diseases
- Death of living organisms
- Loss of vegetation
- Habitat loss
- Ecosystem destruction

Lecture 109

Water pollution:

Water pollution refers to any undesirable change in the quality of water that makes it consumption unfit for human & other animals. Polluted water has serious impact on health of living organisms.

Sources of water pollution:

There are several sources of the water pollution;

Domestic waste, Industrial waste, Untreated sewage, Oil spills, Agricultural runoff, Suspended or particulate matter, Mining activities, & Acid rain.

Industrial and human wastes find their ways into groundwater, streams, lakes, and oceans. When they do, water becomes unfit for human consumption and unfit for wildlife.

Human activities are main cause of water pollution. Water pollution is contaminating all water bodies like lakes, rivers, oceans, aquifers and groundwater day by day.

Lecture 110

Air pollution:

Befouling of air due to release of certain pollutants released due to human activities is known as **Air pollution**. Air pollution also presents serious problems that are fatal to human and other living things.

Air pollutants can be in the form of,

- ❖ Solid particles
- ❖ Gases
- ❖ Liquid droplets

Kinds of air pollutants:

Carbon monoxide(CO), Sulphur dioxide (SO2), Nitrogen dioxide (NO2), Lead, Particulate matter, & Chlorofluorocarbon (CFC) etc.

Effects of Air pollution:

- Acid rain
- Greenhouse effect
- Global warming
- Ozone layer destruction

Acid rain or deposition:

Burning fossil fuels releases sulfur dioxide & nitrogen oxides into the atmosphere. Sulfur dioxide & water combine to produce sulfuric acid, which falls as acid deposition. Acid deposition lowers the pH of lakes.

Greenhouse effect:

CO_2 released from burning fuels accumulates in the atmosphere & prevent the escape of heat coming from earth. This effect is termed as **Greenhouse effect.**

Global warming:

CO_2 reflects solar radiation back to the earth. This reflection of solar radiation is predicted to cause an increase in world temperature, polar ice caps to melt, and ocean levels to rise.

Ozone layer destruction:

The release of chlorinated fluorocarbons from aerosol cans, air conditioners, and refrigerators contributes to the depletion of ozone layer (the earth's ultraviolet filter). It will increase the chance of skin cancer.

When wastes and poisons enter food webs, organisms at the highest trophic levels usually suffer the most.

The accumulation of matter in food webs is called **biological magnification.**

Lecture 111

Noise pollution:

Transportation (cars, trains, airplanes, shipping) and industry (construction or factory) are the leading sources of noise pollution.

Animals rely on hearing to,

1. Communicate 2. Avoid predators 3. Obtain food

Effects of noise pollution,

- Hearing loss
- Loss of communication
- Physiological effects because of increased heart rate and metabolism

Noise from sonar or explosives causes marine mammals to dive deeper. On long, deep dives, more nitrogen enters the blood from the lungs in the form of bubbles and kills the animal.

In humans, this illness is known as the **bends.**

To avoid noise, wildlife may alter their behavior, possibly leaving critical habitat or forage areas, though responses will vary with the kind of noise & the species.

Waterfowl, for example, are particularly disturbed by low-flying aircraft.

Lecture 112

Climate change:

The global climate has changed throughout geologic time on timescales that vary from a few years to millions of years. Some short term global changes occur during volcanic eruptions.

In 1991 Mount Pinatubo in the Philippines erupted, producing huge quantities of gas and ash reaching the stratosphere. It blocked some of the solar radiation from reaching the earth's surface, resulting in a worldwide cooling.

It is estimated that 20,000,000 tons of SO_2 gas erupted into the stratosphere. The resulting H_2SO_4 (sulfuric acid) reflected 2 to 4 percent of the incoming solar radiation back into the space, reducing the mean global temperature at the surface about 0.5°C.

Now a days greenhouse gases have increased the greenhouse effect in the atmosphere due to which the temperature of earth is increasing day by day. This phenomenon is known as **Global warming**.
Human activities are the main cause of this event to occur.

Lecture 113

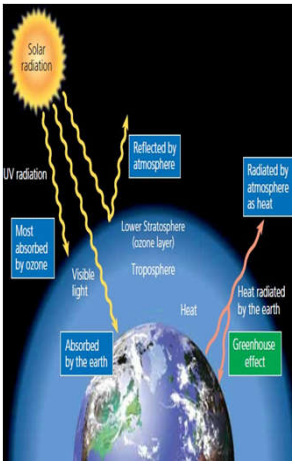
Greenhouse effect:

Small amounts of certain gases, including water vapor (H2O), (CO2), (CH4), & (N2O), in the atmosphere play a role in determining the earth's average temperatures & its climates.

These **greenhouse gases** allow mostly visible light & some infrared radiation and ultraviolet (UV) radiation from the sun to pass through the atmosphere.
The earth's surface absorbs much of this solar energy & transforms it to longer-wavelength infrared radiation (heat), which then rises into the lower atmosphere.

Some of this heat escapes into space, but some is absorbed by molecules of greenhouse gases & emitted into the lower atmosphere as even longer-wavelength infrared radiation.

Some of this released energy radiates into space, and some warms the lower atmosphere & the earth's surface. This natural warming effect of the troposphere is called the **greenhouse effect.**



Causes of Greenhouse effect: Human activities such as, Burning fossil fuels, Deforestation, Growing crops release carbon dioxide, methane, and nitrous oxide into the atmosphere.

Lecture 114

El-Nino effect:

Another type of short-term climate change is the so-called El Nino effect, which occurs every 2 to 7 years around Christmas time, when the typical weather pattern in the Pacific Ocean breaks down.

The trade winds weaken, & low pressure establishes itself in the central Pacific. It causes the winds to blow into the Pacific Basin from the west.
This change causes warm surface water to move toward South & Central America, reversing the directions of some currents & placing warm water along the coasts.

The warm water produces heavy rain over the coastal deserts, subjecting them to disastrous flooding & erosion. The western United States also experiences major flooding events as a result of El Nino.

1. This temporary influx of rain on arid & semiarid regions creates the conditions for disastrous down slope movements in the form of mud slides, destroying entire villages.
2. One result of this temporary climate change is the occurrence of major brush fires, which destroy not only vegetation but also wild and domestic animals.
3. Shifts in warm water in the southern Pacific and Indian Oceans cause cooler water to move along the coast of Australia, resulting in decreased rainfall.

4. The warm water El Nino brings to the coasts of North & South America substantially reduces the upwelling of cold, deep water, adversely affecting the cold water fisheries.

Lecture 115

Global warming:

The average temperature of earth surface is increasing day by day due to increase in concentration of greenhouse gases in the atmosphere. Human activities are main cause of it.

The earth is gradually heating up due to rise in temperature. This phenomenon is called **Global warming**. It is causing serious problems to the health of plants & animals & also the surrounding environment.

Causes of Global warming

Burning of fossil fuels, Volcanic eruption, Industrial effluents, Deforestation, Transportation,& Power plants

Effects of Global warming

Rise in temperature, Rise in Sea Level, Climate change, Extinction, Droughts, & Loss of biodiversity.

Rise in Temperature: The intense heat waves & rising temperatures are becoming more common as greenhouse gases are trapped in the atmosphere. The greenhouse effect thus leads to a rise in temperature, & as it becomes stronger, more heat is trapped within the planet.

Rise in Sea Level: Warm surface temperatures cause glaciers, polar ice to melt. This in turn increases the amount of water in the world's oceans thus contributing to a rise in sea levels. It threatens populations of coastal areas because of their higher vulnerability to flooding.

Climate change:

Gradual increase in temperature has also severely influenced the climate pattern of the earth surface. El nino effect is an example of changing climate due to Global warming.

Extinction:

The polar bear is considered to be an endangered species whose numbers are falling because of their inability to adapt to the volatile temperature changes in the Polar Regions.

Lecture 116

Effects of Global warming:

Some major effects are

- Rise in temperature
- Rise in Sea Level
- Climate change
- Extinction
- Droughts
- Loss of biodiversity

Droughts: A warmer climate will cause shortage of water supply & ultimately crop failure. If these water shortages are persistent it will cause a lot of disruptions in global food production by affecting agriculture & thus leading to situations such as starvation .

Loss of biodiversity: Global warming causes drastic & irreversible changes both in the upper atmosphere & within the planet thus affecting it's every component including land, water, air etc. It makes the organisms unable to survive in their native environments.

Lecture 117

Ozone layer:

A protective layer in the atmosphere that protects the life on earth from harmful effects of ultraviolet radiations from sun is known as Ozone layer. It contains high concentrations of **Ozone** (O_3).

On the base of production ozone is divided into two types,

- **Stratospheric ozone**
- **Tropospheric ozone**

Tropospheric ozone:

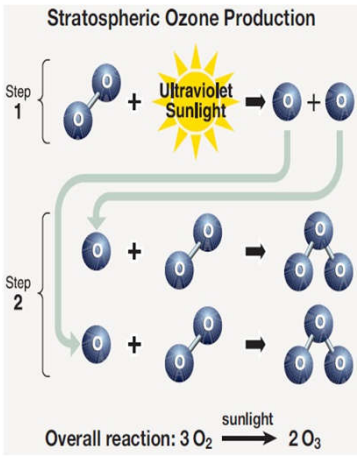
Nitrogen compounds released from automobiles is broken down by sunlight into nitrogen oxide (NO_2). It is then converted in to nitrogen monoxide (NO), releasing an oxygen molecule that combines with oxygen(O_2) in the atmosphere to form **Ozone** (O_3).

Ozone abundances in the stratosphere and troposphere are determined by,

Amount of reactive gases, Sunlight intensity, & Temperature.

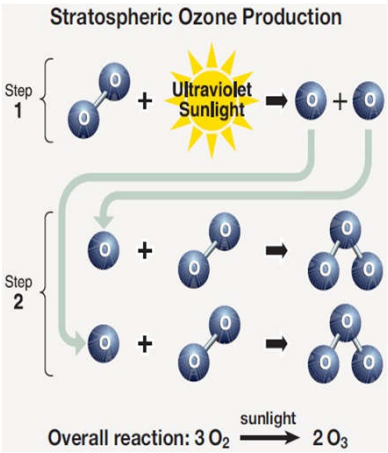
Stratospheric ozone:

In the first step, solar ultraviolet radiation breaks apart one oxygen molecule (O_3) to produce two oxygen atoms.



Ozone, is a deadly poison. However, at the higher levels of the atmosphere, ozone performs an essential function. It shields the surface of the earth from ultraviolet (UV) radiation from the Sun. This radiation is highly damaging to organisms, for example, it is known to cause skin cancer in human beings.

In the second step, each of these highly reactive atoms combines with an oxygen molecule to produce an ozone molecule (O_3).



Lecture 118

Ozone-layer destruction:

Ozone gas is continuously formed by the action of UV rays on molecular oxygen, & also degraded into molecular oxygen in the stratosphere. There should be a balance b/w production & degradation of ozone in the stratosphere.

This balance is disrupted due to ozone degradation by **Chlorofluorocarbons (CFCs).** CFCs are released by refrigerants. CFCs discharged in the lower part of atmosphere move upward and reach into stratosphere.

In stratosphere, UV rays act on them releasing chlorine (Cl) atoms. Cl degrades Ozone releasing molecular oxygen. So CFCs have permanent & continuing affects on Ozone levels.

This increase in release of Chlorofluorocarbons (CFCs) has created a large area of thinned ozone layer in Antarctic region, commonly called as the ozone hole.

Effects of Ozone depletion

Skin cancer , Snow-blindness (inflammation of cornea by UV-rays), Cataract, & Low crop production.

Lecture 119

Solid Waste:

Solid wastes refer to everything that goes out in trash. Municipal solid wastes are wastes from homes, offices, stores, schools, hospitals, etc., that are collected and disposed by the municipality.

Solid waste is usually disposed of in landfills. Landfills take up space &, if not properly contained, can leach toxins into the soil & poison groundwater.

In countries with limited space, solid waste is burned at high temperatures that creates very hazardous ash, & pollutes the air with toxic chemicals.

Hospitals generate hazardous wastes that contain disinfectants and other harmful chemicals, & also pathogenic micro-organisms. Such wastes also require careful treatment & disposal. The use of incinerators is crucial to disposal of hospital waste.

Irreparable computers & other electronic goods are known as **electronic wastes (E-wastes).**

E-wastes are buried in landfills or incinerated.

Recycling involves manual participation thus exposing workers to toxic substances present in e-wastes

Solid waste management: All waste that we generate can be categorised into three types,

1. Recyclable
2. Bio-degradable,
3. Non-biodegradable.

Lecture 120

Strategy to Protect Ecosystems:

Protecting & sustaining ecosystems & the biodiversity contained within them than on saving individual species is a burning issue in all over the world.

Instead of preventing premature extinction of species, it is very important & interesting to protect threatened habitats & ecosystem services.

This ecosystems approach generally would employ the following four point plan:

- Map global ecosystems & create an inventory of the species contained in each of them & the ecosystem services they provide.
- Locate and protect the most endangered ecosystems & species, with emphasis on protecting plant biodiversity & ecosystem services.
- Seek to restore as many degraded ecosystems as possible.
- Make development biodiversity-friendly by providing significant financial incentives (such as tax breaks & write-offs) & technical help to private landowners who agree to help protect endangered ecosystems.

Lecture 121

Stabilize Global Population:

For sustainable balance among the world's biodiversity, ecological systems, & humans it is necessary to stabilize human population.

Humans already capture more than a third of the earth's terrestrial productivity & more than half of the world's fresh water. As food production has doubled in the last thirty-five years, we now release more nitrogen & phosphorus into the environment than all natural sources combined. The following is a partial list of some of actions that can helpful in attaining a stable population. Improve the educational & political status of women. When women are intellectually, economically, & politically empowered, they can make decisions about how many children they wish to have.

- Improve the survival & health of children. Low infant mortality rates & improved prospects for healthy children will definitely allow them to run family businesses & carry on family names.
- Provide easy access to family planning resources. Education on spaced births, delayed marriage, breastfeeding, & birth control will allow parents to choose when & whether to start families & how large those families will be.

Collection and Disposal:

Domestic wastes are collected in small bins, which are then transferred to community bins by private or municipal workers. From these community bins, these are collected & carried to the disposable site.

At the site, garbage is sorted out & separated into biodegradable & non-biodegradable materials.

Non-biodegradable materials such as plastic, glass, metal scraps etc. are sent for recycling. Biodegradable wastes are deposited in land fills & are converted into compost.

The waste if not collected in garbage bins, finds its way into the sewers. Some of it is eaten by cattle. Non-biodegradable wastes like polythene bag, metal scraps, etc. if swallowed by cattle can cost their lives.

All domestic wastes should be properly collected & disposed. The poor management causes health problems leading to epidemics due to contamination of ground water. It is specially hazardous for those who are in direct contact with the waste.

Lecture 122

Waste management:

The improper disposal of wastes is one of the major causes of environmental degradation. Therefore, the management of wastes is of prime importance.

Solid waste management: All waste that we generate can be categorised into three types,

1. Recyclable
2. Bio-degradable,
3. Non-biodegradable.

Lecture 123

Introduction-to biodiversity:

The term biodiversity was used by Edward Wilson to describe the combined diversity at all the levels of biological organisation.

Biodiversity is a vital renewable resource.

In our biosphere immense diversity or heterogeneity exists not only at the species level but at all levels of biological organisation ranging from macromolecules within cells to biomes.

Biological diversity, or **biodiversity,** is the variety of all life forms on the earth.

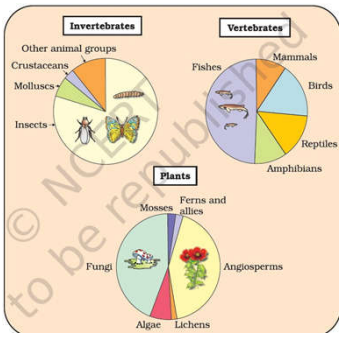
The variety of life on Earth at all its levels, from genes to ecosystems, & the ecological & evolutionary processes that sustain it.

Scientists have identified about 1.8 million species, & every year, thousands of new species are identified.

The identified species include almost a million species of insects, 270,000 plant species, & 45,000 vertebrate animal species.

Among animals, insects are the most species-rich taxonomic group, making up more than 70 per cent of the total. That means, out of every 10 animals on this planet, 7 are insects.

The number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles & mammals.



Biodiversity vary with respect to changes in the following factors,

Climate change, Geology, Natural selection, Species evolution & Sea level.

Components of biodiversity:

Species diversity, Population diversity, Community diversity, Ecosystem diversity, Genetic diversity, Biogeographic diversity & Functional diversity

Biologists are not sure about how many prokaryotic species there might be. The problem is that conventional taxonomic methods are not suitable for identifying microbial species & many species are simply not culturable under laboratory conditions.

Lecture 125

Ecosystem diversity:

Ecosystem diversity refers to variation in the interdependence of biotic communities and the abiotic (non-living) aspects of the environments in which the biotic communities are found.

Lecture 124

Species diversity:

Species diversity refers to number & abundance of species present in different communities.

Variation in the number & phylogenetic diversity (or evolutionary relatedness) of species present in an area.

More than 70 per cent of all the species recorded are animals, while plants (including algae, fungi, bryophytes, gymnosperms & angiosperms) comprise no more than 22 per cent of the total.

The earth's variety of deserts, grasslands, forests, mountains, oceans, lakes, rivers, & wetlands is another major component of biodiversity.

Each of these ecosystems is a storehouse of genetic & species diversity.

An ecosystem is a community of organisms (living things) which depend upon each other for their existence. The variety of processes such as matter cycling & energy flow takes place within ecosystems as species interact with one another in food chains & webs.

Ecosystems can be managed or unmanaged. Most agricultural landscapes are managed ecosystems.

Ecosystems exist at different scales. Interactions between living & non-living things occur at all these scales at the same time.

Ecosystems develop in response to local conditions, which are influenced by

- Climate patterns
- Soil types
- Topography

Because our environment is constantly changing, we need a diverse range of genes to be able to adapt. Preserving variety within populations of species is essential for preserving the ability of that species to cope with environmental change.

Genetic diversity will help the individuals to survive under following conditions of environmental stress,

Climate change, Drought, Fire, famine &Invasive species

Lecture 126

Genetic diversity:

Genetic Diversity is the variety of genetic material within a species or a population. Each species is made up of individuals that have their own particular genetic composition. This means a species may have different populations, each having different genetic compositions.

Genes are the basic units of life on Earth. They are responsible for both the similarities and the differences b/w organisms, e.g. the changes in colour & markings of birds within a particular species.

Genes are simply traits that we inherit from our parents. They can pass on to our children. For example ,traits such as hair colour, eye colour, curly or straight hair & whether you can curl your tongue & so on. Genetic diversity enables life on the earth to adapt to & survive dramatic environmental changes. In other words, genetic diversity is vital to the sustainability of life on earth.

Lecture 127

Types of biodiversity over spatial scale:

Robert Whittaker (1972) created a system to describe biodiversity over different spatial scales.

- Alpha diversity
- Beta diversity
- Gamma diversity

Alpha diversity: It refers to the diversity within a particular area or ecosystem, & it is usually expressed by the number of species in that ecosystem. This is equivalent to measuring the species richness of an area.

For example, if we want to monitor aquatic biodiversity on the offshore side of a fringing coral reef in the Ocean.

Then the alpha diversity of this reef is measured by the total number of species found in the area that we are monitoring.

We can compare the degree of species change as we move along the transect from one ecosystem to the next.

This comparison of so-called between-area diversity is referred to as **Beta diversity.**

Beta diversity is a comparison of diversity between ecosystems, usually measured as the amount of species change b/w the ecosystems.

Thus, beta diversity allows us to compare diversity b/w ecosystems.

Gamma diversity is a measure of the overall diversity within a large region, & so in some respects it is an expanded version of alpha diversity.

Thus, if we chose to survey all the reefs of a coast side of a ocean & measure species diversity over the whole area, then that would be gamma diversity.

We could expand the survey even further to include the reefs of the neighbouring Islands.

The gamma diversity would then include the species for all those islands.

Lecture 128

Importance of biodiversity:

Biodiversity performs important biological services.

- **Regulation of Climate**
- **Soil formation and protection**
- **Water resources protection**
- **Education and scientific value**
- **Recreation and ecotourism**
- **Pollution breakdown and absorption**
- **Economic value**

Regulation of Climate: Undisturbed forests maintain the rainfall in their immediate environment by contributing to the hydrological cycle.

Soil formation and protection: Biodiversity plays an important role in the formation of soil & the maintenance of soil structure & the retention of moisture & nutrient levels.

Water resources protection :

Natural vegetative cover in watersheds protects ecosystems from extreme events such as flood & drought, & regulates & stabilizes water runoff thereby maintaining water cycles & water quality.

Education and scientific value:

The natural environment has been the basis for many books, television programs & movies produced for entertainment & educational purposes, as well as in scientific ecological observations.

Recreation and ecotourism:

The natural environment is used for recreation through activities, such as hiking, bird watching, nature walks & SCUBA diving. Ecotourism, as an industry, is growing rapidly in many developing countries.

Pollution breakdown and absorption:

Ecosystems and ecological processes play an important role in the breakdown & absorption of many pollutants created by humans & their activities. Some such pollutants are garbage sewage & oil spills.

Economic value

- Goods
- Food
- Wood & Forest Products

Lecture 129

Threats to biodiversity:

The biological wealth of our planet has been declining rapidly & the accusing finger is clearly pointing to human activities. The colonization of tropical Pacific Islands by humans is said to have led to the extinction of more than 2,000 species of native birds.

Many species persist on a greatly reduced area of their former range & on increasingly fragmented landscapes.

Ecosystems suitable for agriculture, such as tropical dry forests & tall-grass prairie, have almost completely disappeared from our planet. Dams are disrupting freshwater ecosystems, while the marine world is threatened by overfishing & habitat destruction.

Humans are also transporting plants & animals around globe both deliberately & unintentionally. These "invaders" threaten other species or change entire ecosystems.

Human influence reaches even the farthest corners of the globe; species in the Arctic & Antarctic are contaminated by pollutants created thousands of miles away & carried through the atmosphere.

We are modifying the functioning of the entire planet, changing the earth's atmosphere through,

- Industrial release of carbon dioxide
- Diminishing the ozone layer by the production of chlorofluorocarbons.

In general, loss of biodiversity in a region may lead to,

- ❖ Decline in plant production.
- ❖ Lowered resistance to environmental distresses such as drought.
- ❖ Increased variability in certain ecosystem processes such as plant productivity, water use, & pest & disease cycles.

Main threats to biodiversity are,

Habitat loss & fragmentation, Over-exploitation, Invasive species, Extinction & Pollution.

Lecture 130

Habitat loss & fragmentation:

Habitat loss is the permanent conversion of land to other uses. Habitat loss drives habitat fragmentation, & the two are tightly linked. Fragmentation of habitats is typically a consequence of habitat loss.

Habitat loss & fragmentation is the main threat to wildlife & the primary causes of species extinction. Although habitats can be fragmented or lost because of natural disturbances, such as earthquakes or geological forces but humans are the principal threat.

People have modified landscapes for thousands of years for agricultural production. Today, agricultural activities are the major cause of habitat loss and fragmentation throughout the world.

Human settlement, resource extraction, & industrial development generally result in small, isolated areas or patches of natural habitat surrounded by developed land.

- ➢ Mangroves, sea grasses, salt marshes, coral reefs, kelp forests, & rocky shorelines are fragmented by natural forces such as Wave action & currents.
- ➢ Tides

Storm surge

Human activities such (dredging, boating and nutrient enrichment).

Lecture 131

Invasive species:

Species whose populations have expanded dramatically & out-compete or displace native species, potentially threatening the structure & function of intact ecosystems.

When alien species are introduced unintentionally or deliberately for whatever purpose , some of them turn invasive, & cause decline/extinction of indigenous species.

The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.

The recent illegal introduction of the African catfish *Clarias gariepinus* for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.

A well-known example is the brown tree snake, which was introduced inadvertently on many Pacific islands and subsequently caused the extinction of a number of native birds, bats and lizards.

There appear to be three major stages in the process of invasion by exotic species:

- Dispersal
- Establishment
- Integration

Lecture 132

Overexploitation:

Overexploitation of resources (or using resources at an unsustainable rate) is a critical problem in conservation.

The overexploitation or nonsustainable use of wildlife is closely linked and plays an increasing role in the loss of biodiversity.

Overharvesting, nonsustainable use, & the illegal trade in some species are threatening not only their continued survival but also that of ecosystems & the livelihoods of communities & local economics that depend upon them.

Unsustainable hunting, fishing, logging, or gathering of wild populations leads to their commercial, ecological or global extinction.

Overexploitation can be divided into two major categories:

- **Direct exploitation**
- **Indirect exploitation**

Direct exploitation: Direct exploitation ranges from commercial activities such as logging operations or trade in endangered species to subsistence hunting.

Indirect exploitation:

Indirect exploitation encompasses the unintentional mortality of non target species such as fish or turtles killed as by-catch in fishery operations. Both endanger species around the world.

Lecture 133

Pollution:

Pollution is any undesirable change in physical, chemical or biological characteristics of air, land, water or soil. Agents that bring about such an undesirable change are called as pollutants.

Everyday, thousands of pollutants are discharged into our environment. Many pollutants lack regulation, and their lingering presence threatens biodiversity, affecting individual species or degrading entire ecosystems.

Sulfur and nitrogen oxides are released into the atmosphere when fossil fuels, such as coal in power plants, or oil in vehicles, or wood, are burned. These combine with water in the atmosphere to create sulfuric and nitric acid, which fall to earth as “acid rain”.

Acid rain also dissolves other harmful metals, such as mercury, which plants & animals then absorb. On land, pollution by acid rain & other air pollutants tends to affect plants more than animals.

Chlorinated hydrocarbons, such as the insecticide DDT (dichloro-diphenyl-trichloro-ethane) & PCB (polychlorobenzene), are renowned for their toxic effects on the environment.

A particularly troubling characteristic of these pollutants is their ability to persist over long time frames and spread over large areas. A phenomenon called bio magnification is well known for mercury and DDT.

A few toxic substances, often present in industrial waste waters, can undergo biological magnification (Bio magnification) in the aquatic food chain.

Bio magnification refers to increase in concentration of the toxicant at successive trophic levels.

Lecture 134

Deforestation:

Deforestation is the conversion of forested areas to non-forested ones.

According to an estimate, almost 40 per cent forests have been lost in the tropics, compared to only 1 per cent in the temperate region.

One of the major reasons is the conversion of forest to agricultural land so as to feed the growing human population. Trees are axed for timber, firewood, cattle ranching & for several other purposes.

Effects of deforestation:

- Increase in CO_2 conc. in the atmosphere.
- Habitat destruction.
- Climate change.
- Loss of biodiversity.
- Hydrologic cycle disturbances.
- Soil erosion.
- Desertification.

Reforestation is the process of restoring a forest that once existed but was removed at some point of time in the past. It is basically a rehabilitation process of forests.

Reforestation may occur naturally in a deforested area.

However, we can speed it up by planting trees with due consideration to biodiversity that earlier existed in area.

Lecture 135

Resource depletion & biodiversity:

Humans are either directly or indirectly exploiting about 40% of the earth's net primary production. It involves converting natural areas to agricultural uses, frequently substituting less efficient crop plants for native species.

The main threats to biological diversity arise from habitat destruction by expanding human populations. Habitat loss displaces thousands of native plants & animals.

Some of the most important threatened natural areas include,

- Tropical rain forests,
- Coastal wetlands,
- Coral reefs.

Tropical rain forests have probably received the most attention. Tropical rain forests cover only **7%** of the earth's land surface, but they contain more than **50%** of the world's species.

Tropical rain forests are being destroyed rapidly, mostly for agricultural production. About 76,000 km^2 (an area larger than Costa Rica) is being cleared each year.

Clearing of tropical rain forests achieves little, because the thin, nutrient-poor soils of tropical rain forests are exhausted within two years.

Tropical Deforestation: Forests are often cut & burned to make land available for agriculture.

Sadly, rain forests are a nonrenewable resource. Seeds of rain forest plants germinate rapidly, but seedlings are unprotected on sterile, open soils. Even if a forest were able to become reestablished, it would take many centuries to return to a climax rain forest.

Characteristics of the Five Kingdoms

Characters	Five Kingdoms				
	Monera	**Protista**	**Fungi**	**Plantae**	**Animalia**
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellulosic (Polysaccharide + amino acid)	Present in some	Present (without cellulose)	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation	Cellular	Cellular	Multiceullar/ loose tissue	Tissue/ organ	Tissue/organ/ organ system
Mode of nutrition	Autotrophic (chemosyn-thetic and photosynthetic) and Hetero-trophic (sapro-phytic/para-sitic)	Autotrophic (Photosyn-thetic) and Hetero-trophic	Heterotrophic (Saprophytic/ Parasitic)	Autotrophic (Photosyn-thetic)	Heterotrophic (Holozoic/ Saprophytic etc.)

Lecture 136

Five-kingdom classification:

In Linnaeus' time a **Two Kingdom** system of classification with **Plantae** & **Animalia** kingdoms was developed that included all plants & animals respectively.

This system did not distinguish between,

- Eukaryotes & prokaryotes,
- Unicellular & multicellular organisms.
- Photosynthetic (green algae) & non-photosynthetic (fungi) organisms.

R.H. Whittaker (1969) proposed a Five Kingdom classification.

1. Monera 2. Protista 3. Fungi

4. Plantae 5. Animalia.

The main criteria used for classification,

- Cell structure
- Thallus organization
- Mode of nutrition
- Reproduction
- Phylogenetic relationships.

Lecture 137

Kingdom-Monera: Concept:

Bacteria are the sole members of the **Kingdom Monera**. They are the most abundant micro-organisms. Bacteria occur almost everywhere. Hundreds of bacteria are present in a handful of soil.

They also live in extreme habitats such as,

- Hot springs
- Deserts
- Snow
- Deep oceans

Bacteria are grouped in 4 categories on their shape,

1. Coccus 2. Bacillus 3. Vibrium 4. Spirillum

Coccus:

These are spherical shape bacteria.



Bacillus:

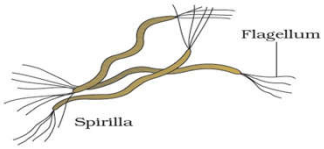
These are rod shaped bacteria.



Vibrium: These are comma shaped bacteria.



Spirillum:



These are spiral shaped bacteria.

There are two groups of bacteria on the base of food manufacturing,

1. Autotrophic bacteria
2. Heterotrophic bacteria

Autotrophic bacteria are further divided into two groups,

- Photosynthetic autotrophic bacteria
- Chemosynthetic autotrophic bacteria

Methanogens:

These are found in swap & marshy areas. Methanogens (*Methanobacterium*) are present in the gut of several ruminant animals such as cows & buffaloes & they are responsible for the production of methane (biogas) from the dung of these animals.

Archeobacteria differ from other bacteria in having a different cell wall structure and this feature is responsible for their survival in extreme conditions.

Lecture 138

Archeobacteria:

The Archeobacteria are the most primitive life-forms known on earth. All members of the Archeobacteria inhabit anaerobic environments. These environments may reflect the conditions on the earth at the time of life's origin.

These bacteria are special since they live in some of the most harsh habitats.

1. Halophiles
2. Thermophiles
3. Psychrophiles
3. Methanogens

Halophiles: These types of bacteria are found in extreme salty areas. They are found in great salt lake and dead sea.

Salinibacter ruber is an example of this kind of bacteria.

Thermophiles: These types of bacteria are found in hot springs areas(usually in deep sea vents called hydrothermal vents). ***Thermus aquaticus*** & ***Thermococcus litoralis*** are example of this group.

Psychrophiles: These are also named as cold loving bacteria and are found in Antarctic and Arctic region. ***Listeria monocytogenes*** is an example of this group.

Lecture 139

Eubacteria:

There are thousands of different **eubacteria** or 'true bacteria'. They are characterized by the presence of a rigid cell wall, & if motile, a flagellum is present in them.

Food synthesis: On the base of food synthesis bacteria are of two types,

1. Autotrophic bacteria **2. Heterotrophic bacteria**

1. Autotrophic bacteria:

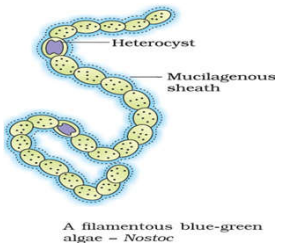
- ❖ Photosynthetic autotrophs
- ❖ Chemosynthetic autotrophs

❖ **Photosynthetic autotrophs:**

The **cyanobacteria** (also referred to as blue-green algae) have chlorophyll *a* similar to green plants & are **photosynthetic autotrophs** .The cyanobacteria are unicellular, colonial or filamentous, freshwater/marine or terrestrial algae.

Some of these organisms can fix atmospheric nitrogen in specialized cells called **heterocysts**. Examples are,

Nostoc Anabaena



A filamentous blue-green algae – *Nostoc*

❖ **Chemosynthetic bacteria:**

These bacteria oxidize various inorganic substances such as nitrates, nitrites & ammonia & use the released energy for their ATP production. They play a great role in recycling nutrients like N, P, Fe & S.

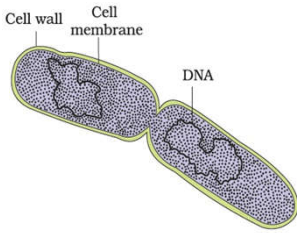
2. Heterotrophic bacteria:

Some characters of these bacteria are,

1. Decomposers 2. Synthesize curd from milk
3. Antibiotic production 4. Nitrogen fixation
5. Pathogens

Bacterial reproduction:

Bacteria reproduce mainly by fission. Sometimes, under unfavorable conditions, they produce spores.



A dividing bacterium

Mycoplasma:

The Mycoplasma are bacteria that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen. Many mycoplasma are pathogenic in animals and plants.

Mode of reproduction:

Protists reproduce asexually by the process of cell fusion & sexually by a process involving zygote formation.

Classification of protists:

1. Chrysophytes
2. Dinoflagellates
3. Euglenoids
4. Slime molds
5. Protozoans

Examples of protists:

- Amoeba
- Paramecium
- Euglena
- Diatoms
- Dinoflagellates

Lecture 141

Chrysophytes:

Chrysophytes are plant like protists. They are mostly unicellular or colonial organisms which are found in fresh & salt water throughout the world.

Examples of Chrysophytes:

1) Diatoms 2) Golden brown algae
3) Yellow-green algae

Characteristics of Chrysophytes:

1. Unicellular eukaryotes 2. Photosynthesis
3. Freshwater & saltwater habitat 4. Planktons

Lecture 140

Kingdom-Protista: Concept:

Members of the kingdom **Protista** are eukaryotic and consist of single cells or colonies of cells.

Protista was first used by **Ernst Haeckel** in the year 1886. This kingdom forms a link between other kingdoms of plants, animals and fungi.

Characteristics of protists,

1. Aquatic habitat 2. Unicellular eukaryotes
3. Flagella or cilia 4. Cyst formation
5. Autotrophic or heterotrophic

Diatoms:

1. Shells with silica 2. Photosynthesis
3. Major producers in oceans 4. Water filters
5. Diatomaceous earth

Diatomaceous earth:

The shells of fossil diatoms form thick deposits on the sea floor known as diatomaceous earth. Being gritty this soil is used in polishing, filtration of oils and syrups.

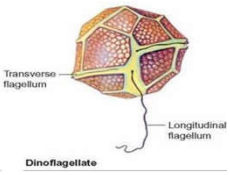
Lecture 142

Dinoflagellates:

Dinoflagellates are prominent members of both the phytoplankton and the zooplankton of marine and freshwater ecosystems. They are also common in benthic environments and in sea ice.

Characters of Dinoflagellates:

- Mostly marine and photosynthetic. Photosynthetic Dinoflagellates have yellowish or brownish plastids (pigment-containing bodies) and may store food in the form of starches, starch like compounds, or oils.
- Most of them have two flagella; one lies longitudinally and the other transversely in a furrow between the wall plates.



- They appear yellow, green, brown, blue or red depending on the main pigments present in their cells.
- The cell wall has stiff cellulose plates on the outer surface.
- Although sexual processes have been demonstrated in a few genera, reproduction is largely by binary or multiple fission.
- Under favorable conditions Dinoflagellates populations may reach 60 million organisms per liter of water.

Red tide: Red Dinoflagellates (Example: *Gonyaulax)* undergo such rapid multiplication that they make the sea appear red.



Some Dinoflagellates produce toxins that are among the most poisonous known. Toxins released by such large numbers may even kill fish and other marine animals.

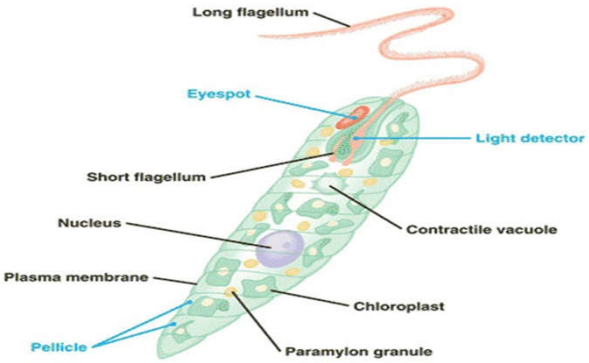
Lecture 143

Euglenoids:

Organisms called Euglenoids, belongs to algal phylum Euglenophyta in the kingdom Protista. They make up a large group of common microorganisms whose known species range is between 750-900 species.

Characters of Euglenoids:

- Majority of them are fresh water organisms found in stagnant water.
- Instead of a cell wall, they have a protein rich layer called pellicle which makes their body flexible.
- They have two flagella, a short and a long one.



- A small light sensitive eye spot is present.
- Though they are photosynthetic in the presence of sunlight, when deprived of sunlight they behave like heterotrophs by predating on other smaller organisms.
- Interestingly, the pigments of Euglenoids are identical to those present in higher plants.
- They have both plant and animal-like features, which makes them difficult to classify and hence they are called as connecting link between plants and animals.

Lecture 144

Slime moulds:

Slime moulds are a kind of protists that aggregates into colonies and ingest fungi, bacteria, protozoa, other micro-organisms, and dead & decaying organic matter.

Slime molds are divided in two taxonomic groups,

1. Cellular slime molds (Phylum Acrasiomycota)
2. Plasmodial slime molds (Phylum Myxomycota)

Organisms in both groups are eukaryotic.

Characters of slime moulds:

- Slime moulds are saprophytic protists.
- The body moves along decaying twigs & leaves engulfing organic material.
- Under suitable conditions, they form an aggregation called plasmodium which may grow & spread over several feet.
- They are extremely resistant and survive for many years, even under adverse conditions.
- During unfavorable conditions, the plasmodium differentiates and forms fruiting bodies bearing spores at their tips. The spores possess true walls.
- The spores are dispersed by air currents.

Lecture 145

Protozoans:

All protozoans are heterotrophs and live as predators or parasites.

1. Amoeboid protozoans 2. Flagellated protozoans
3. Ciliated protozoans 4. Sporozoans

❖ **Amoeboid protozoans:**

These organisms live in fresh water, sea water or moist soil. They move & capture their prey by putting out pseudopodia (false feet) as in *Amoeba*. Marine forms have silica shells on their surface. Some of them such as *Entamoeba* are parasites.

❖ **Flagellated protozoans:**

- Free-living or parasitic. They have flagella.
- The parasitic forms cause diastases such as sleeping sickness. Example: ***Trypanosoma***.

❖ **Ciliated protozoans:**

- Aquatic and movement by cilia.
- A cavity (gullet)opens to outside of the cell surface.
- The coordinated movement of rows of cilia causes the water loaded with food to be driven into the gullet.
- Example: ***Paramoecium***

Sporozoans:

This includes diverse organisms that have an infectious spore-like stage in their life cycle. The most notorious is *Plasmodium* (malarial parasite) causes malaria disease which has a staggering effect on human population.

Lecture 146

Kingdom-fungi: concept:

The fungi constitute a unique kingdom of heterotrophic organisms. They show a great diversity in morphology & habitat. Fungi are cosmopolitan & occur in air, water, soil & on animals & plants.

Examples of fungi:

1. Mushrooms 2. Penicillium 3. Yeast 4. Puccinia

Structure of fungi:

- The body of fungus consists of long, slender thread-like structures called **Hyphae**.

- The network of hyphae is known as **Mycelium**.
- Some hyphae are continuous tubes filled with multinucleated cytoplasm these are called **Coenocytic hyphae**.
- Others have septae or cross walls in their hyphae.
- The cell walls of fungi are composed of chitin and polysaccharides.

Mode of nutrition:

- Saprophytes
- Parasites
- Symbionts

Fungi classification:

Kingdom fungi is classified into 4 groups on the base of,

1. The morphology of the mycelium.
2. Mode of spore formation and fruiting bodies.

Four groups of fungi:

1. Phycomycetes
2. Ascomycetes
3. Basidiomycetes
4. Deuteromycetes

- **Saprophytes:** Most fungi are heterotrophic & absorb soluble organic matter from dead substrates & hence are called **saprophytes.**
- **Parasites:** Those that depend on living plants and animals are called **parasites**.
- **Symbionts:** They can also live as **symbionts** – in association with algae as **lichens** and with roots of higher plants as **mycorrhiza**.

Mode of reproduction:

1) Sexual reproduction
2) Asexual reproduction

Lecture 147

Phycomycetes:

Members of Phycomycetes are found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites on plants.

Characters of Phycomycetes:

- The mycelium is aseptate and coenocytic.
- Asexual reproduction takes place by zoospores (motile) or by aplanospores (non-motile).
- These spores are endogenously produced in sporangium.

Sexual reproduction:

1. Oospores 2. Ascospores 3. Basidiospores

Asexual reproduction:

Asexual reproduction is by spores called conidia or sporangiospores or zoospores.

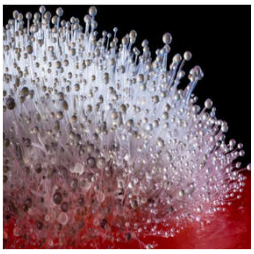
Reproduction cycle:

1. Fusion of protoplasms between two motile or non-motile gametes called **plasmogamy.**
2. Fusion of two nuclei called **karyogamy.**
3. Meiosis in zygote resulting in haploid spores.

- A zygospore is formed by fusion of two gametes.
- These gametes are similar in morphology (isogamous) or dissimilar (anisogamous or oogamous).

Examples of Phycomycetes:

- *Mucor*
- *Rhizopus* (the bread mould)
- *Albugo* (the parasitic fungi on mustard).



Mucor spp. growing on a damaged tomato, showing fruiting bodies.

Lecture 148

Ascomycetes:

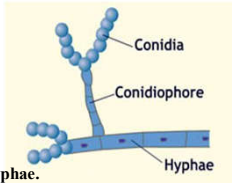
Ascomycetes are commonly known as sac-fungi. The Ascomycetes are mostly multicellular, e.g., *Penicillium*, or rarely unicellular, e.g., yeast (*Saccharomyces*).

Characters of Ascomycetes:

- Saprophytic.
- Decomposers.
- Parasitic or coprophilous (growing on dung).
- Mycelium is branched and septate.
- The asexual spores are conidia produced exogenously on the special mycelium called conidiophores. Conidia on germination produce mycelium.
- Sexual spores are called ascospores which are produced endogenously in sac like asci (singular ascus). These asci are arranged in different types of fruiting bodies called ascocarps.

Examples of Ascomycetes:

1. *Aspergillus* 2. *Claviceps*
3. *Neurospora.*



Ascomycete with septate hyphae.

Neurospora is used extensively in biochemical and genetic work. Many members like morels and truffles are edible and are considered delicacies.

Lecture 149

Basidiomycetes:

Commonly known forms of basidiomycetes are mushrooms, bracket fungi or puffballs.

They grow in soil, on logs and tree stumps and in living plant bodies as parasites, e.g., rusts and smuts.

Characters of Basidiomycetes:

- The mycelium is branched and septate.
- The asexual spores are generally not found, but vegetative reproduction by fragmentation is common.
- The sex organs are absent, but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes. The resultant structure is dikaryotic which ultimately gives rise to basidium.
- Karyogamy and meiosis take place in the basidium producing four basidiospores. The basidiospores are exogenously produced on the basidium (pl.: basidia).
- The basidia are arranged in fruiting bodies called basidiocarps.

Examples of Basidiomycetes:

- *Agaricus* (mushroom)
- *Ustilago* (smut)
- *Puccinia* (rust fungus)



Agaricus (Mushroom)

Lecture 150

Deuteromycetes:

Commonly known as imperfect fungi because only the asexual or vegetative phases of these fungi are known. When the sexual forms of these fungi were discovered they were moved into classes they rightly belong to.

When the linkages were established, the fungi were correctly identified and moved out of deuteromycetes.

Once perfect (sexual) stages of members of dueteromycetes were discovered they were often moved to ascomycetes and basidiomycetes.

Characters of Deuteromycetes:

- The deuteromycetes reproduce only by asexual spores known as conidia.
- The mycelium is septate and branched.
- Some members are saprophytes or parasites while a large number of them are decomposers of litter and help in mineral cycling.

Examples of **Deuteromycetes:**

1. *Alternaria* 2. *Colletotrichum* 3. *Trichoderma*.